IBM Blue Vela SSS6000- 32 GPFS nodes

MLPerf Storage v2.0 Submission

System name: IBM Blue Vela

Submitter name: Brian Belgodere

Submission date: July 11, 2025

Version of the benchmark: MLPerf Storage v2.0

Solution type: Shared filesystem using a proprietary access protocol

Submission division: CLOSED

Power Requirements: 533040W

Table of Contents

IBM Blue Vela SSS6000 - 32 GPFS nodes *1*

1. Hardware *3*

1.1. Overview **3**

1.2. Client Server **4**

1.3. Storage Server **5**

1.4. Network **6**

1.5. Power and Rack Units Requirements **7**

2. Software *8*

2.1. Client Servers **8**

2.2. Storage Servers **8**

2.3. Network Switches **8**

3. Settings *9*

3.1. Compute Server **9**

3.1.1. Scale Tunings 9

3.2. Storage Server **9**

3.2.1. Scale Tunings 9

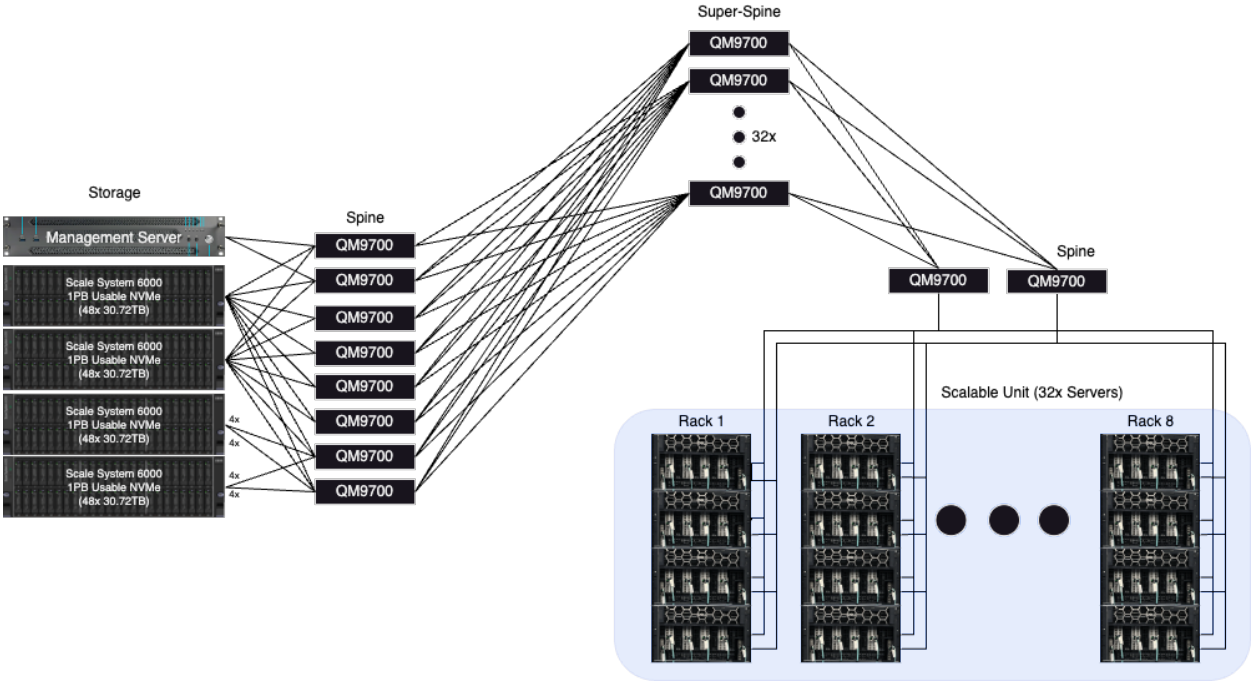
3.2.2. Filesystem Configuration 11

3.3. Network Switch **11**

1. Hardware

1.1. Overview

The hardware used consisted of 32x client servers connected through a spine, super-spine network architecture to the IBM Scale System Storage. The benchmarked storage and compute clients are part of a larger AI training cluster IBM Blue Vela which is described in greater detail here: https://arxiv.org/pdf/2407.05467

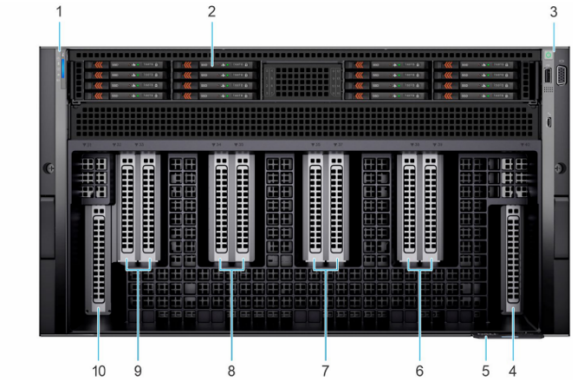


1.2. Client Server

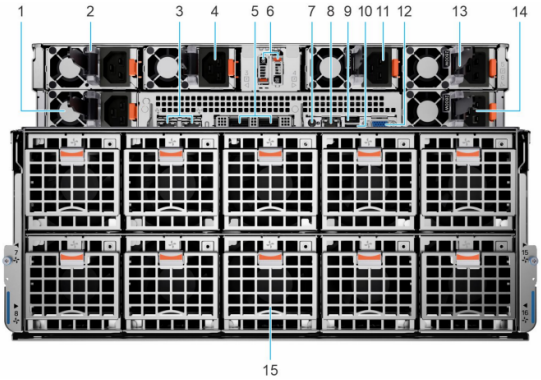
The clients used are Dell XE9680 HGX compute servers (32x) with the following specs:

- 6U Server
- Dual 48-core 4th Gen Intel Xeon (8468) Scalable Processors
- 2TB DDR5 ECC memory 4800MT/s (32x 64GB)
- 8x NVIDIA H100 GPUs with 80GB High Bandwidth Memory (HBM)
- 10x NVIDIA ConnectX-7 NDR 400 gigabits per second (Gb/s) InfiniBand Host Channel Adapters (HCA)
    - 8 dedicated to compute fabric
    - 2 dedicated to storage fabric
- 2x 480GB NVMe U.2 Gen4 (boot)
- 8x 3.4TB Enterprise NVMe U.2 Gen4
- Dual 25G Ethernet Host links
- 1G Management Ethernet Port

Front View:



Rear View:



Front View
- (1) Left control panel
- (2) Drives
- (3) Right control panel
- (4) PCIe slot 40
- (5) Information tag
- (6)-(9) PCIe slots
- (10) PCIe slot 31

Rear View
- (1,2,4,10,12,13) PSU
- (3) LOM NIC
- (5) OCP NIC
- (6) BOSS-N1
- (7) System Identification button
- (8) iDRAC9 ethernet
- (9) USB 2.0
- (11) VGA
- (14) Fan module
- (15) ???

1.3. Storage Server

The storage servers (4x) used are IBM Storage Scale System 6000 (SSS6000) solutions with the following specs:

- 4U Storage Server – Dual IO canisters (2U)
- Dual AMD EPYC Genoa 48C CPU per IO Canister
- 1.5TB Memory total – 768GB per IO canister
- 48x 30.72TB PCIe Gen4 NVMe SSD
- 8x NVIDIA ConnectX-7 NDR 400 gigabits per second (Gb/s) InfiniBand Host Channel Adapters (HCA) - 4x per IO canister

In addition to the Storage Server, there is a management server with the following specs:

- 2U Management Server
- Single AMD EPYC 16C CPU
- 128GB Memory
- 1x NVIDIA CX-7 Dual-port 200Gb VPI HCA
- 1x 4-Port 10GbE NIC

Storage Front

Storage Rear

Note: Benchmark system highlighted in green – ESS3500 NOT used

1.4. Network

The network topology is more complex than required for the benchmark as the systems used for the benchmark are part of a much larger AI training cluster. The networking is NDR InfiniBand utilizing 42x NVIDIA Mellanox QM9700 switches deployed in a spine, super-spine architecture supporting 24x total NVIDIA scalable units (SU) which is outside of the scope of the benchmark however notable due to the perceived complexity of the network topology.

Storage Switches

Compute Switches

Note: There is a separate compute InfiniBand fabric NOT used in benchmark tests, Storage switches highlighted in green.

1.5. Power and Rack Units Requirements

Power requirements table.

System component	Power Supply Unit	Nameplate rated power	Design power
DLIO client	Power Supply 1	2800W	14000W
	Power Supply 2	2800W	
	Power Supply 3	2800W	
	Power Supply 4	2800W	
	Power Supply 5	2800W	
	Power Supply 6	2800W	
32x DLIO clients – total		**537600W**	**448000W**
Storage Shelf (2 IO controllers)	Power Supply 1	2400W	3200W
	Power Supply 2	2400W	
	Power Supply 3	2400W	
	Power Supply 4	2400W	
4x Storage Shelfs – total		**38400W**	**12800W**
Storage Management Servers	Power Supply 1	2000W	1200W
	Power Supply 2	2000W	
Switch	Power Supply 1	747W	1720W
	Power Supply 2	747W	
42x Switches – total		**62748W**	**72240W**
Totals		**642748W**	**534240W**

Rack Units table

System component	Rack Units per component	Number of components	Totals
DLIO client	6	32	192
Storage Shelf (2 IO controllers)	4	4	16
Storage Management Server	2	1	2
Switch	1	42	42
Totals			**252**

2. Software

2.1. Client Servers

The Dell XE9680 HGX Servers are running the following software:

OS:	RedHat Enterprise Linux (RHEL) 9.5
Kernel:	5.14.0-503.23.1.el9_5.x86_64
OFED:	MLNX_OFED_LINUX-24.10-1.1.4.0 (OFED-24.10-1.1.4)
IBM Scale:	5.2.2.1

2.2. Storage Servers

The Scale System 6000 servers are running the following software:

Scale System:	6.2.2.0 (default)
OS:	RedHat Enterprise Linux (RHEL) 9.4
Kernel:	5.14.0-427.42.1.el9_4.x86_64
OFED:	MLNX_OFED_LINUX-24.04-0.6.6.5 (OFED-24.04-0.6.6)
IBM Scale:	5.2.2.0 efix3 (340711)

2.3. Network Switches

The NVIDIA Mellanox QM9700 InfiniBand switches are running the following software:

Firmware:	31.2012.3040

3. Settings

3.1. Compute Server

The IBM Storage Scale (GPFS) client software v5.2.2.1 is installed on the compute nodes, and they are part of a Scale “Client Cluster” which remotely mounts the Scale filesystem from the “Storage Cluster” which consists of the Scale System 6000s. The filesystem is accessed through the standard Scale (GPFS) NSD protocol as-if they were part of one larger cluster.

3.1.1. Scale Tunings

Some scale client tunings were applied, all non-default values are listed below:

```
[root@p1-r29-n1 ~]# mmdiag --config | grep !
! ccrEnabled 1
! clusterId 14605646151343901334
! clusterName dfw.bluevela.rmf.ibm.com
! dmapiFileHandleSize 32
! ignorePrefetchLUNCount 1
! maxblocksize 16777216
! maxFilesToCache 131072
! maxStatCache 131072
! minReleaseLevel 3300
! myNodeConfigNumber 164
! numaMemoryInterleave yes
! pagepool 34359738368
! proactiveReconnect 0
! sdrNotifyAuthEnabled yes
! tscCmdAllowRemoteConnections no
! verbsPorts mlx5_1/1 mlx5_6/1
! verbsRdma 1
! verbsRdmaCm disable
! verbsRdmaSend 1
! workerThreads 1024
```

3.2. Storage Server

The default deployment method for Scale System 6.2.2.0 was used.

3.2.1. Scale Tunings

The default Scale System 6000 tunings were applied as part of initial installation. All non-default values either from installation scripts (default) and/or customizations are listed below:

```
[root@ess6000io1 ~]# mmdiag --config |grep !
! ccrEnabled 1
! clusterId 9941035806433427289
! clusterName dfw_ess6000.bluevela.rmf.ibm.com
! dmapiFileHandleSize 32
! ignorePrefetchLUNCount 1
! maxblocksize 16777216
! maxFilesToCache 131072
```

! maxMBpS 50000
! maxStatCache 131072
! minReleaseLevel 3300
! myNodeConfigNumber 1
! nsdMaxWorkerThreads 3842
! nsdMinWorkerThreads 3842
! nsdRAIDBlockDeviceMaxSectorsKB 0
! nsdRAIDBlockDeviceNrRequests 0
! nsdRAIDBlockDeviceQueueDepth 0
! nsdRAIDBlockDeviceScheduler off
! nsdRAIDClientOnlyChecksum 1
! nsdRAIDDefaultGeneratedFD 0
! nsdRAIDDiskDiagTimeout 130
! nsdRAIDEventLogToConsole all
! nsdRAIDMasterBufferPoolSize 2147483648
! nsdRAIDReconstructAggressiveness 0
! nsdRAIDSmallThreadRatio 2
! nsdRAIDSSDPerformanceShortTimeConstant 2500000
! nsdRAIDThreadsPerQueue 16
! nsdRAIDTracks 327680
! nsdSmallThreadRatio 1
! nspdBufferMemPerQueue 25165824
! nspdQueues 240
! nspdThreadsPerQueue 2
! numaMemoryInterleave yes
! pagepool 486136610816
! pagepoolMaxPhysMemPct 90
! panicOnIOHang 1
! pitWorkerThreadsPerNode 32
! prefetchPct 50
! sdrNotifyAuthEnabled yes
! tscCmdAllowRemoteConnections no
! verbsPorts mlx5_0/1 mlx5_1/1 mlx5_2/1 mlx5_3/1
! verbsRdma 1
! verbsRdmaCm disable
! verbsRdmaSend 1
! workerThreads 1024

3.2.2. Filesystem Configuration

The filesystem gpfs0_bench was created using default configuration parameters with the details outlined below:

```
v6000ems-vm[root@ems2-vm1 ~]# mmlsfs gpfs0_bench
flag value description ------------------- ------------------------ -----------------------------------
-f                 16384                    fragment (subblock) size in bytes
 -i                4096                     Inode size in bytes
-I                 32768                    Indirect block size in bytes
-m                 1                        Default number of metadata replicas
-M                 2                        Maximum number of metadata replicas
-r                 1                        Default number of data replicas
-R                 2                        Maximum number of data replicas
-j                 scatter                  Block allocation type
-D                 nfs4                     File locking semantics in effect
-k                 all                      ACL semantics in effect
-n                 1024                     Estimated number of nodes that will mount file system
 -B                16777216                 Block size
 -Q                none                     Quotas accounting enabled
                   none                     Quotas enforced
                   none                     Default quotas enabled
--perfileset-quota no                       Per-fileset quota enforcement
--filesetdf        no                       Fileset df enabled?
-V                 33.00 (5.1.9.0)          File system version
--create-time      Thu Jul 3 22:43:40 2025  File system creation time
-z                 no                       Is DMAPI enabled?
-L                 33554432                 Logfile size
-E                 yes                      Exact mtime mount option
-S                 relatime                 Suppress atime mount option
-K                 whenpossible             Strict replica allocation option
--fastea           yes                      Fast external attributes enabled?
--encryption       no                       Encryption enabled?
--inode-limit      150003712                Maximum number of inodes
--uid              07647F0A:6867079C        File system UID
--log-replicas     0                        Number of log replicas
--is4KAligned      yes                      is4KAligned?
--rapid-repair     yes                      rapidRepair enabled?
--write-cache-threshold 0                   HAWC Threshold (max 65536)
--subblocks-per-full-block 1024             Number of subblocks per full block
-P                 system                   Disk storage pools in file system
--file-audit-log   no                       File Audit Logging enabled?
--maintenance-mode no                       Maintenance Mode enabled?
--flush-on-close   no                       flush cache on file close enabled?
--auto-inode-limit no                       Increase maximum number of inodes per inode space automatically?
--nfs4-owner-write-acl yes                  NFSv4 implicit owner WRITE_ACL permission enabled?
 --inode-segment-mgr no                     Inode Segment Manager enabled?
–d
RG001LG001VS004;RG001LG002VS004;RG001LG003VS004;RG001LG004VS004;RG002LG001VS004;RG002LG002VS004;RG002LG003VS004;RG002LG004VS004;RG0
03LG001VS004;RG003LG002VS004;RG003LG003VS004;
-d                 RG003LG004VS004;RG004LG001VS004;RG004LG002VS004;RG004LG003VS004;RG004LG004VS004 Disks in file system
-A                 yes                      Automatic mount option
 -o                none                     Additional mount options
-T                 /gpfs/gpfs0_bench        Default mount point
--mount-priority   0                        Mount priority
```

3.3. Network Switch

No special network tuning applied to QM9700 switches – default configuration